

LOG6310E - Engineering Digital Twin

Final Report

Orthogonal Turning Digital Shadow

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Abstract

This report presents the design and implementation of an Orthogonal Turning Digital Shadow developed to monitor machining stability and support decision-making during cutting operations. Orthogonal turning is widely used in precision manufacturing, particularly in aerospace applications where expensive materials, strict tolerances, and high-quality requirements make process stability essential. One of the main challenges in this context is chatter, a self-excited vibration phenomenon that degrades surface quality, increases tool wear, reduces productivity, and may ultimately damage the workpiece or machine. To address this problem, the project implements a microservice-based digital shadow that receives process data, transforms it into meaningful stability indicators, and communicates the current machining condition through analytical and visual interfaces. Because the system was not connected to the real lathe during this phase of development, synthetic data generated from a one-degree-of-freedom (1DOF) Simulink model was used to emulate stable and unstable cutting scenarios. The architecture integrates multiple services for data production, FFT-based frequency analysis, stability evaluation, prediction, process reconfiguration, formal verification, time-series logging, dashboard reporting, and 3D visualization in Godot. RabbitMQ is used as the communication backbone, while Influx DB supports historical storage and querying. Using Kritzinger's framework, the project can be best classified as a digital shadow rather than a digital model or a full digital twin. The system goes beyond a digital model because it includes automatic data flow from the represented process to the virtual environment, but it does not yet achieve the bidirectional, validated real-machine integration required of a true digital twin. The project demonstrates that a modular digital shadow can provide near-real-time insight into machining stability, support corrective reasoning, and establish a strong foundation for future integration with the physical turning system. Challenges such as connectivity and validation were experienced during the duration of this project. On another note, lessons learned regarding the multidisciplinary requirements of the digital twin field were further understood and the agile methodology was used to meet with domain experts, present progress and perform validation checks. In terms of limitations, the existing digital shadow currently only applies for a specific orthogonal turning process, however, future work is underway to test the microservice architecture using real data, generalize the framework and further develop services such as the prediction service.

Keywords: Digital Shadow, Aerospace Machining, Stability, Vibrations, Chatter, Fast-Fourier Transform (FFT)

Introduction

Orthogonal turning is a machining process in which a cutting tool removes material from a rotating workpiece under controlled cutting conditions. This specific cutting method requires the cutting tool to be perpendicular to the cutting direction [1]. It is representative of many industrial turning operations and is especially relevant in aerospace manufacturing, where parts are often produced from high-value materials such as aluminum, titanium, and nickel-based alloys. In these contexts, dimensional accuracy, surface integrity, and process stability are critical because even small deviations during cutting can reduce part quality, accelerate tool wear, increase defect rates and overall increase production cost while reducing productivity. A major challenge in turning operations is the dynamic interaction between the tool, workpiece, and machine structure. As cutting conditions change, vibration levels may increase and eventually lead to chatter, a self-excited and chaotic instability that degrades machining performance [2]. Because chatter can emerge quickly and evolve during the cut, monitoring the process through a conventional trial-and-error approach is inefficient and risky, particularly when working with expensive materials and tight tolerances.

To address this challenge, this project team developed an Orthogonal Turning Digital Shadow that monitors process behavior in near real-time and provides actionable insight on machining stability. The proposed system combines synthetic machining data, microservice-based communication, signal analysis, visualization, and decision-support capabilities to represent the state of the cutting process in the virtual domain. By doing so, the project team aims to support safer and more informed operation of the machining system while laying the groundwork for a future transition from a digital shadow to a more fully connected digital twin.

Problem Statement

The central problem addressed in this project is the onset of unstable vibrations during orthogonal turning. When the cutting process operates outside the stability region, vibration amplitudes can grow and turn into fully developed chatter. This phenomenon is one of the most harmful and well-known problems in machines, since it affects both process quality and productivity. Chatter can produce a poor surface finish, reduce dimensional accuracy, damage the workpiece, shorten tool life, and, in severe cases, force the cutting process to stop altogether. Notably, chatter is the number one biggest productivity offender in the aerospace machining industry [3]. In aerospace and other high-precision manufacturing domains, instability's consequences are particularly significant. The materials involved are costly, the acceptable tolerance ranges are small, and the production environment demands repeatable quality. A single unstable cut may therefore lead not only to scrap material, but also to lost machine time, increased tooling cost, and can eventually damage the machines. For this reason, maintaining cutting conditions within the stability region is essential.

The difficulty lies in the fact that instability is not always obvious from direct observation alone and usually once you can observe it, it's usually too late. The machining process generates multiple signals, such as acceleration, force, displacement, and sound, and the clear indicators of instability may only become clear after appropriate analysis. This creates a need for a system

that can receive process data, transform it into meaningful stability indicators, and present the results in a form that supports timely decision-making in near real-time.

Project Motivation

The motivation for this project is to provide a practical digital engineering solution for monitoring and understanding machining stability during orthogonal turning. The digital shadow will eventually be able to make changes in cutting parameters, namely spindle speed (n_s), during cutting to maintain stability. It currently only changes the behavior of visualization. Rather than relying only on offline analysis or operator intuition, the proposed digital shadow offers a structured way to collect, process data, analyze vibration and its behavior, and communicate the current state of the system through user-facing interfaces. This is valuable because it makes the evolution of stability more visible as it occurs, allowing earlier identification of unsafe or inefficient operating conditions. A second motivation is the problem's multidisciplinary nature. Monitoring machine stability is governed by both physical dynamics and software-system design. As a result, an effective solution must combine knowledge from manufacturing, vibration analysis, data processing, communication infrastructure, visualization and much more. In this project, that integration is achieved through a microservice-based architecture in which different services handle synthetic data, First-Fourier Transform-based (FFT) signal transformation, stability assessment, physical twin reconfiguration, prediction, reporting through a dashboard, formal verifications of processes, and 3D visualization. This modular design improves maintainability, scalability, and clarity of responsibilities across the system [4].

Finally, the project is motivated by the longer-term vision of moving from observation to intelligent intervention. In its current form, the system primarily functions as a digital shadow that monitors the process and provides insight into whether cutting remains stable. However, it also includes the foundation for virtual-to-physical feedback by generating reconfiguration decisions, such as adjusting spindle speed or triggering an emergency stop under unstable conditions. This capability demonstrates the practical value of the system: it does not merely display data but helps determine what should happen next to protect the workpiece, the tool, and the machining process.

Methodology

The development of the Orthogonal Turning Digital Shadow followed an iterative and multidisciplinary methodology combining domain knowledge, synthetic data generation, software architecture design, signal-processing analysis, and user-facing visualization. The first step was to increase the team's understanding of the physical system under study: an orthogonal turning operation on a lathe. To ensure that digital representation addressed a real industrial need, the team consulted domain experts in machining and stability analysis throughout the development of the solution. These discussions helped refine the project's scope and, more importantly, guide the selection of stability indicators that would be meaningful for monitoring chatter and unstable cutting behavior. Due to time and safety constraints, the project was not connected to the physical machine, rather synthetic data was generated to imitate incoming data from the physical twin. A one-degree-of-freedom (1DOF) dynamic model in MATLAB's Simulink was created by domain experts to simulate machining behavior under stable and unstable cutting conditions. This approach allowed the team to create repeatable scenarios and evaluate how instability evolves over time without requiring immediate access to a fully instrumented experimental setup. The synthetic data included variables such as acceleration, displacement, and process parameters like width of cut (b) and spindle speed (n_s), which were then used as the input stream for the digital shadow pipeline.

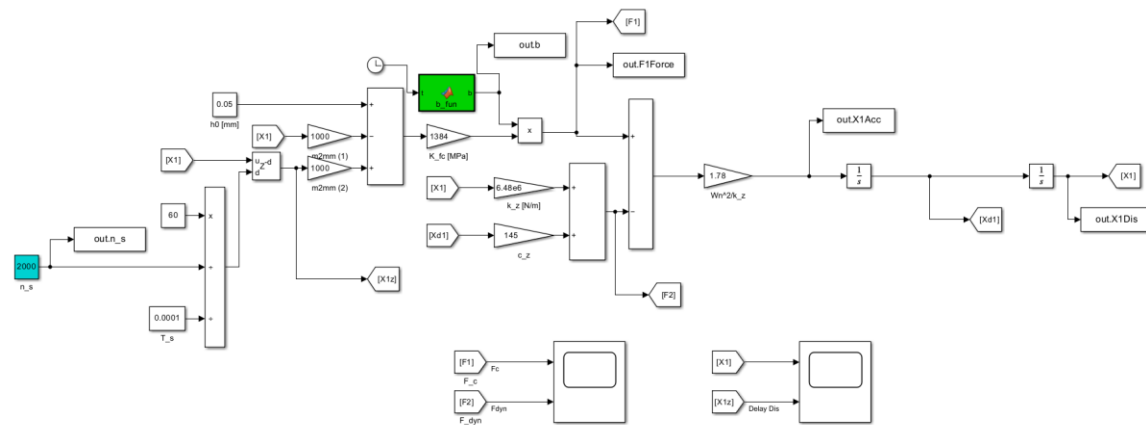


Figure 1 Simulink 1 Degree of Freedom Model

Once the input data source was established, the system was designed as a microservice-based architecture. This methodological choice was made to improve modularity, maintainability, and clarity of responsibilities across the project. Instead of building a single monolithic application, the digital shadow was decomposed into smaller services for data production, frequency analysis, stability evaluation, process reconfiguration, dashboard communication, visualization, and data logging. RabbitMQ served as the message broker for asynchronous communication between services, while Python implemented core analytics and orchestration logic. In addition, Docker was used to allow the team to containerize the solution, MATLAB's Simulink was used for synthetic data generation, GODOT gaming engine was used for the 3D visualization and finally

GitHub was used for versioning purposes. This structure also made it easier to test components independently and to evolve the project incrementally over time.



Figure 2 Tools Used to Build Digital Shadow Solution

For process monitoring, the methodology emphasized frequency-domain analysis rather than relying only on raw time-domain signals. Based on feedback from domain experts, the team adopted (FFT) analysis to convert acceleration measurements into dominant frequency and amplitude indicators. The evolution of the vibration amplitude was then used to determine whether the process was stable, approaching instability, or clearly unstable. On top of this, a decision layer was added to support process reconfiguration, such as reducing spindle speed by 10% when warning conditions were confirmed or issuing an emergency stop under severe instability. In parallel, prediction and verification capabilities were introduced to enhance the system's usefulness and trustworthiness. The prediction service estimates future signal behavior over a short horizon, while formal verification services check whether warning-confirmed, emergency-stop and process-completion actions are correctly reflected in the visualization layer.

The methodology also included the design of user-facing services so that the digital shadow would not only compute results but also communicate them effectively. A dashboard was developed to display live data, FFT-based indicators, status updates, and prediction outputs, while a Godot-based 3D visualization was used to intuitively represent the machining process and its operational state. The dashboard also logs all the data coming in each second and this log is made available at the end of the process. In addition, Influx DB services were incorporated to support time-series logging, querying, and testing of the pipeline. Together, these components allowed the project to provide both machine-oriented analytics and human-oriented insight.

Kritzinger's Framework

According to Kritzinger's framework, the Orthogonal Turning solution is best classified as a digital shadow. In Kritzinger's classification, a digital model is a digital representation of a physical system without automatic data exchange between the physical and virtual domains. Updates may be manual, offline, or only loosely connected. The present project goes beyond that level because its virtual environment is continuously and automatically driven by incoming process data. Synthetic machining data, movement information, and cutting parameters are published through the producer and RabbitMQ pipeline, then automatically consumed by the FFT, stability-check, prediction, dashboard, logging, and Godot visualization services. For this reason, the system is not merely a static or manually updated model. The project is classified as a digital shadow because it includes automatic, ongoing data flow from the process side to the virtual side. The digital environment does not need manual intervention to update its state; instead, it receives process data, transforms it into stability indicators, visualizes the machining condition, and supports control-oriented reasoning in near real time. This one-way automatic synchronization

from the represented process to the digital system is the key feature that distinguishes a digital shadow from a digital model.

At the same time, the system is not yet a full digital twin. Although the architecture can generate actions such as spindle-speed reduction and emergency-stop commands, these actions are not yet validated through trusted bidirectional integration with the real physical turning machine. The current implementation relies mainly on simulated data and software-level feedback, and the generated actions primarily affect the digital environment and visualization rather than a fully instrumented physical lathe operating under real cutting conditions. To reach the level of a true digital twin, the system would require validated real-machine sensing, trusted physical actuation, and experimental verification in closed-loop operation. This classification reflects the maturity of the project accurately: the system clearly exceeds the level of a digital model because it is automatically driven by incoming process data, but it remains short of a full digital twin because its bidirectional interaction with the real physical process is not yet fully realized.

Architecture

The architecture of the Orthogonal Turning Digital Shadow was designed as a modular service-based system that separates data generation, analytics, decision support, visualization, and formal verification into distinct but connected components. This structure improves maintainability, scalability, and testing, while also making the relationships among the system's main elements easier to understand. At the architecture level, the system can be described through four main aspects: models, data sources, enablers, and interactions. The main **model** foundation is the simulation-based representation of the turning process used to generate synthetic machining data (1DOF). Additional analytical models are then applied throughout the pipeline, including frequency-domain analysis through FFT, trend-based stability evaluation, and short-horizon prediction. Together, these models progressively transform raw process signals into stability indicators and decision-support outputs.

The main **data source** in the current implementation is the synthetic dataset produced from the simulation model. This data includes acceleration, movement, and cutting parameters such as spindle speed and width of cut. These signals are injected into the architecture through the producer service, which acts as the entry point for the rest of the system. Although the current implementation relies on simulated data, the architecture is designed so that real machine signals could later replace this source with minimal structural changes.

The system is supported by several **enablers**. RabbitMQ provides the communication backbone that allows asynchronous interaction between services. Python is used to implement the core service logic and analytics. Godot supports the 3D visualization of the cutting process, Dash and Plotly support the dashboard interface, and Influx DB provides time-series storage and querying. These enablers are essential because they enable the architecture to connect analysis, decision-making, persistence, and user interaction into a single coherent pipeline.

The main **interactions** in the architecture follow a staged flow. The producer service publishes displacement, acceleration, and cutting parameter data. The FFT service consumes acceleration data and extracts dominant frequency and amplitude information. The stability-check

service then evaluates the vibration trend and classifies the cutting process state. This classification is passed on to the PT reconfiguration service, which determines whether the system should remain stable, issue a warning, reduce spindle speed, or trigger an emergency stop. In parallel, the dashboard and visualization services receive system information and present it to the user in graphical and 3D form, while Influx DB-related services support storage and later analysis of system behavior. Formal verification services monitor key events and verify that important actions are correctly reflected across downstream components.

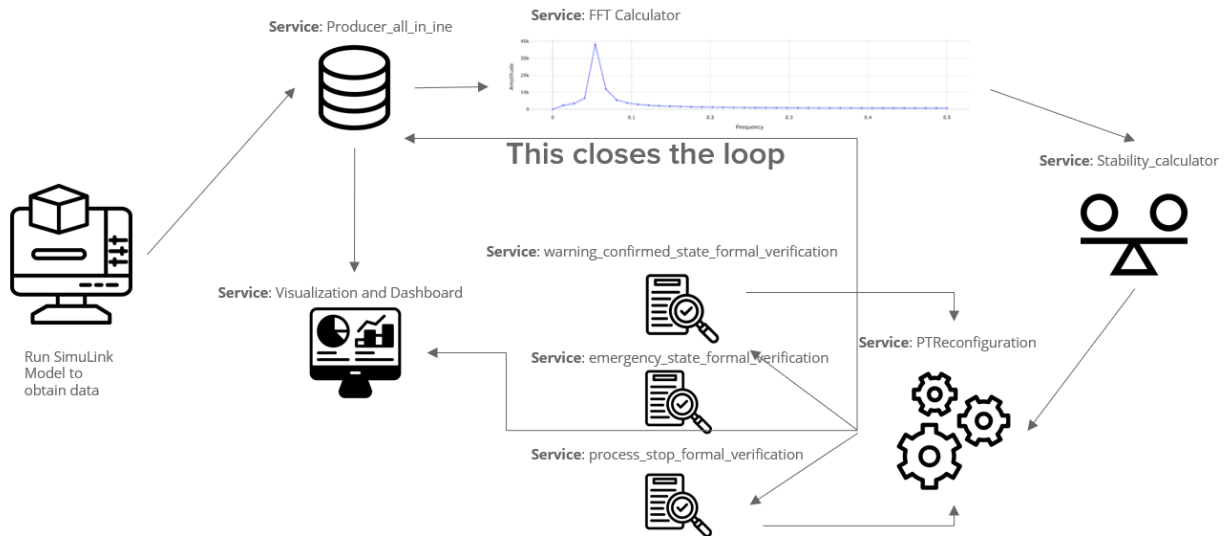


Figure 3 Digital Shadow Solution

Diagrams

The diagrams developed for this project are important because they present the digital shadow from complementary perspectives. While the architecture section explains the system in terms of models, data sources, enablers, and interactions, the diagrams visually clarify the system's operational logic and flow. In this project, the most important diagrams are the state machine, microservice and constellation diagrams. The state machine diagram explains how the machining process evolves from one operating state to another, while the microservice and constellation diagram explains how data and decisions move through the digital shadow pipeline.

State Machine Diagram

The state machine diagram represents the main operational states of the orthogonal turning process and the transitions between them. The process begins in the **Idle** state, where the machine is not yet cutting. Once the workpiece and tool are properly set up, the system transitions into **Stable Cutting**. This state represents normal operation and is further refined into two substates, **Impact**, and **Cutting** because of the orthogonal cutting geometry, to distinguish the initial tool-workpiece contact from the subsequent continuous cutting phase. This distinction is useful because the first contact produces a transient response that differs from steady cutting behavior.

As the machining process continues, the digital shadow monitors the vibration response and evaluates the stability condition of the cut. If the measured behavior suggests that instability is increasing, the process transitions from **Stable Cutting** to the **Warning state**. The warning state corresponds to a condition in which the process is no longer comfortably stable, but corrective action may still return it to acceptable operation. In the logic of this project, this corrective action is represented by a spindle-speed adjustment intended to push the process back toward the stability region. If the adjustment is successful, the system returns from **Warning** to **Stable Cutting**, indicating that the instability was detected early and the corrective action was sufficient. If the adjustment is not successful and the FFT amplitude continues to trend upward, the system transitions to **Emergency Stop**. This state represents a severe instability condition in which the machining process must be stopped to avoid damage to the workpiece, tool, or machine. Once the process is stopped or the cut is completed, the system returns to **Idle**.

This diagram is valuable because it clearly expresses the operational reasoning of the digital shadow. It shows that the system not only observes the machining process but also interprets its condition and provides different levels of response based on the severity of the detected instability.

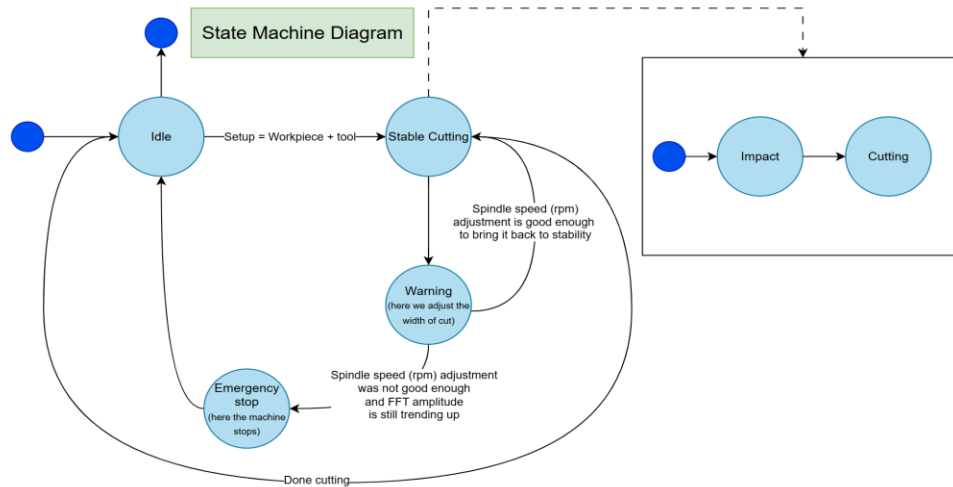


Figure 4 State Machine Diagram

Diagram of Microservices

The microservice diagram presents the digital shadow as an interconnected architecture organized around the relationship between the physical twin (PT) and the digital environment (DT). On the PT side, the system includes the lathe system, the PT simulator, and the cutting parameters, such as the width of cut and spindle speed. On the DT side, the architecture contains communication, analytics, storage, verification, visualization, and decision-support services that process and interpret machining information. At the center of the architecture is RabbitMQ, which serves as the message broker for asynchronous communication between services. Instead of coupling services directly, RabbitMQ allows each component to publish and consume information independently. This makes the architecture modular and easier to scale, while allowing multiple services to react to the same data stream simultaneously. In this project, RabbitMQ distributes simulated process data to analytics, database, dashboard, verification, and visualization services.

The analytical path begins with the incoming simulated machining data through the main **producer service**. The **FFT service** converts time-domain vibration signals into the frequency domain, making dominant vibration behavior easier to identify. The **Stability Calculator** then processes these outputs and determines whether the process is stable or unstable. In parallel, the **Prediction Service** uses processed signal history to estimate near-future behavior, thereby extending the system from reactive monitoring toward predictive support. The architecture also includes components for data persistence and traceability. The **Database Recorder** captures incoming measurements and stores them in Influx DB, which functions as the system's time-series database. This allows the project to preserve historical information for later querying, analysis, and dashboard use. These components are important because they extend the system beyond live monitoring and support retrospective study of machining behavior over time.

Decision-making and feedback are handled by the **PT Reconfiguration service**. This service receives output from the analytical components and determines whether to adjust the cutting parameters. In the current implementation, it can issue corrective actions such as reducing spindle speed (ns) or triggering an emergency stop. This service provides the main feedback link from the digital environment toward the physical process and therefore represents the strongest control-oriented element of the architecture.

Three formal verification services, **Warning Confirmed**, **Emergency State**, and **Process Completion** are also included in the diagram. Their role is to verify that critical decisions generated by the PT Reconfiguration service are correctly reflected across downstream components, such as the visualization service. This increases confidence in the system's reliability by verifying that important warnings and emergency actions are not only generated but also consistently propagated throughout the rest of the pipeline.

Finally, the user-facing components of the architecture are represented by the **Dashboard** and the Godot-based **Visualization service**. The dashboard presents charts, indicators, and status information, while the 3D visualization provides a more intuitive representation of the machining process and its changing condition. Together, these services make the outputs of the digital shadow observable and interpretable to the user.

Table 1 Microservice Inputs and Outputs

Service	Input	Output
Data Producer / Simulation Service	CSV / Excel Simulation files	raw acceleration (accel_chunks), tool position (movement_queue), process context (b_mm, ns_rpm) - all published to RabbitMQ
Database Recording Service	All 6 RabbitMQ queues: accel_chunks, fft_peak, stability_trend, godot_status, movement_queue, process_context	InfluxDB measurements: accel_chunk, fft_peak, stability, pt_status, movement, process_params, run_session
Database Query Service	Influx DB time-series data	Pandas DataFrames; amplitude forecasts (linear regression); similar episode matches (cross-correlation); unified event replay timeline
Frequency Calculator Service (FFT)	Raw acceleration chunk (accel_chunks)	Peak vibration frequency (f_peak_hz) and amplitude (amp_peak) published to fft_peak
Stability Calculator Service	FFT peak amplitude (fft_peak)	Trend classification (UP / DOWN / FLAT / MIXED) over the last 3 amplitudes, published to stability_trend
PT Reconfiguration Service	Stability trend (stability_trend) + process parameters (process_context)	Machine status (STABLE / WARNING / EMERGENCY_STOP) → godot_status; control commands (UPDATE_NS, EMERGENCY_STOP) → pt_actions; verification events →

		verification_events_warning, verification_events_emergency
RabbitMQ-to-Godot Bridge Service	movement_queue, godot_status, process_context from RabbitMQ; state reports from Godot via WebSocket	WebSocket messages to Godot; Godot state republished to godot_state_warning_verifier, godot_state_emergency_verifier
Visualization Service (Godot)	WebSocket messages from bridge (position, status, spindle speed)	3D cutting simulation rendered in real time; state reports sent back to bridge
Warning Formal Verification Service	verification_events_warning (WARNING_CONFIRMED event); Godot state from godot_state_warning_verifier	Pass / Fail log confirming Godot spindle speed matches commanded value
Emergency Formal Verification Service	verification_events_emergency (EMERGENCY_STOP event); Godot state from godot_state_emergency_verifier	Pass / Fail log confirming Godot is frozen; re-issues EMERGENCY_STOP to pt_actions if check fails
Dashboard Service	All analytics results + control commands	Real-time web UI (Dash / Plotly) — display only, no output

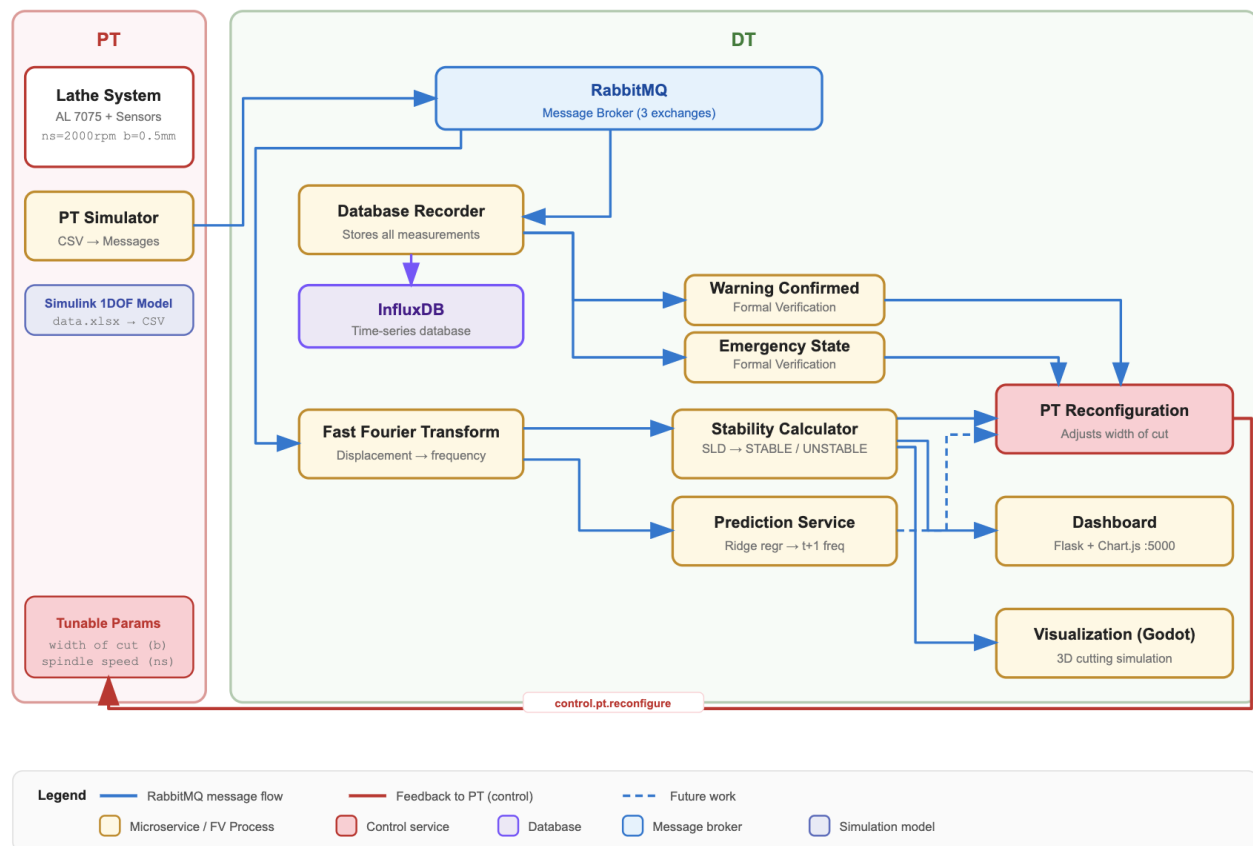


Figure 5 Diagram of Microservices

Microservices

The Orthogonal Turning Digital Shadow is implemented as a collection of specialized microservices, each responsible for a distinct part of the monitoring, analysis, visualization, or decision-support pipeline. This decomposition was chosen to improve modularity, maintainability, and testability, while also allowing the system to evolve incrementally as the project matured. Instead of placing all functionality in a single application, the project separates data production,

analytics, prediction, persistence, visualization, and verification into independent services that communicate through RabbitMQ. The following subsections describe the role, inputs, outputs, and value of each implemented microservice.

1- Producer_all_in_one (Producer)

It is the entry point of the digital shadow pipeline. Its main role is to publish the simulated process data used by the rest of the system. Every second, this service publishes one thousand points to RabbitMQ for the other services to use. It loads movement data for visualization, acceleration data for analysis, and cutting parameter information such as spindle speed (ns) and width of cut (b). It then publishes these messages to the appropriate RabbitMQ queues at a controlled rate so that downstream services can process them as if they were receiving a live stream. In addition to data publication, this producer also plays an orchestration role. It starts key analysis services, maintains the current process context, and listens for control actions such as spindle-speed updates or emergency-stop commands generated elsewhere in the system. This makes it more than a simple data sender: it acts as the operational source of the simulated physical process. The value of this service lies in its ability to emulate live machining conditions and provide a controlled, repeatable input stream for testing the rest of the digital shadow.

2- Influx_logger_service

As our digital shadow pipeline runs, data is constantly moving between services in the form of vibration readings, frequency peaks, stability judgements, machine status updates, tool positions, and cutting parameters. The Database Recording Service listens to all these data streams simultaneously and quietly saves everything to Influx DB as it arrives. It does not interfere with any of the processing; it simply observes and records. To handle the volume of incoming data without latency and errors, it runs six listeners at the same time, one for each type of data stream. Each listener is responsible for a specific category of information, and when a new reading arrives, it saves a structured snapshot of that reading to the database, tagged with the time it was recorded and the session it belongs to. This tagging is important: every simulation run is given a unique identifier, so historical records from different experiments can later be separated and compared without confusion. The six listeners each cover a distinct part of the pipeline. The first captures raw acceleration chunks from the sensor stream, saving a statistical summary of each chunk including its average, spread, and peak values, rather than every individual sample. The second records the dominant vibration frequency and its strength at each time step, as produced by the FFT service. The third saves the stability trend judgement, whether vibration is rising, falling, flat, or mixed, along with the three amplitude values that led to that conclusion. The fourth records every machine status decision made by the reconfiguration service, including the current cutting parameters at the time of that decision. The fifth logs the tool position data that drives the 3D visualization. The sixth captures the cutting process parameters, spindle speed (ns) and width of cut (b), as they are reported by the data producer throughout the simulation.

3- Influx_query_service:

While the recording service handles writing data in, the Database Query Service handles reading it back out. It provides a way for other parts of the system like the dashboard to ask questions about what has happened over time. A user can request data from any measurement over any time window, filter it by experiment session, or ask for a summarized view where readings are averaged across larger time intervals rather than shown point by point. The service is also able to pull together the FFT readings, stability judgements, and machine status decisions from a past session and line them all up in a single chronological view. This makes it straightforward to trace exactly what the system saw, concluded, and decided at each moment during a run.

4- FFT_service

It performs the main frequency-domain transformation in the digital shadow. It consumes acceleration chunks, one thousand points, published by the producer and applies FFT analysis to identify the dominant vibration frequency and its amplitude. These outputs are then published for use by the next stage of the analytical pipeline. This service is central to the project because the team's methodology shifted toward frequency-domain analysis as the most meaningful way to monitor instability progression. Rather than relying only on raw time-domain vibration signals, the FFT service extracts more informative indicators of dynamic behavior. Its value lies in converting raw acceleration measurements into features that support chatter detection and stability reasoning. This service publishes the coordinate of (frequency, amplitude) with the highest amplitude to the stability check service.

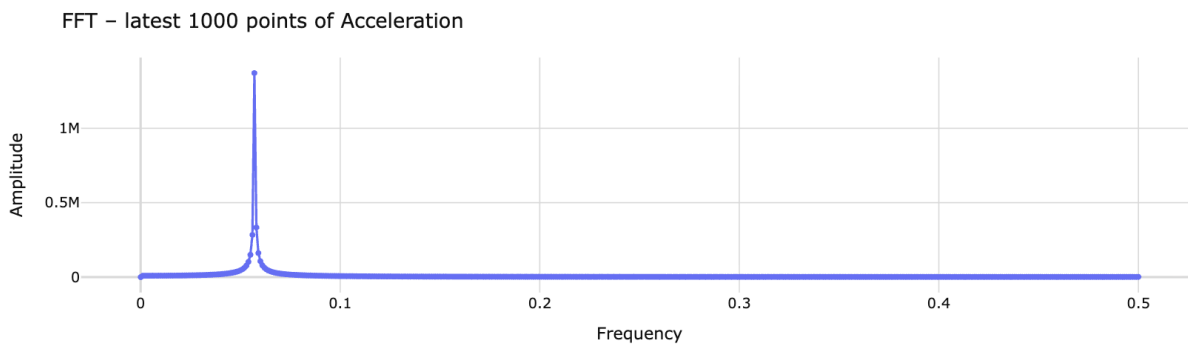


Figure 6 FFT Diagram, Amplitude as a Function of Frequency

5- Stability_check_service

It consumes the outputs of the FFT service and evaluates the recent trend of the vibration amplitude using the last 3 points or the last 3 seconds. By examining how the peak amplitude evolves over successive samples, it determines whether the process trend is increasing, decreasing, flat, or mixed. These classifications are then published as stability-trend outputs for downstream decision making. The importance of this service lies in its interpretation of FFT results. The FFT service provides frequency-domain indicators, but the stability-check service transforms those indicators into a process-level judgment about whether the machining condition

is moving toward instability. In other words, it converts signal features into operational meaning. This makes it a key decision-support component of the digital shadow.

6- Prediction_model (Prediction Service)

It is the first step to extend the system from reactive monitoring toward predictive capability. It processes signal history and derived features to estimate the short-term future behavior of the monitored process. In the current implementation, the service uses envelope extraction, peak tracking, and regression-based prediction to forecast how the vibration trend may evolve over a short horizon. This service is valuable because it will eventually allow the system to anticipate instability rather than only respond after it becomes visible in the current signal. Estimating what may happen next provides the user with a more proactive form of decision support. Even though prediction remains an evolving part of the project and currently only acts as a monitoring tool, it demonstrates how the digital shadow can move toward more intelligent and forward-looking behavior.

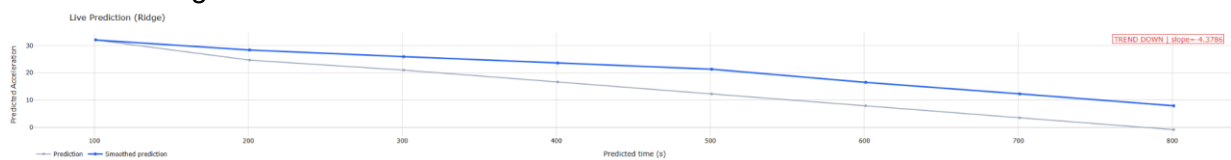


Figure 7 Prediction Service - Stable Pass

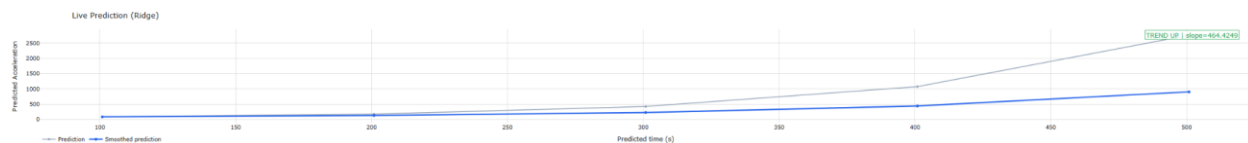


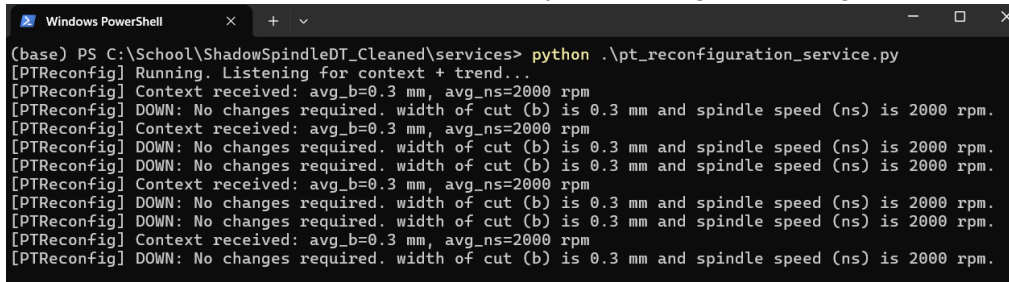
Figure 8 Prediction Service - Unstable Pass

These two figures show the result of the prediction service in two different cases the first plot shows the result of prediction when the machine is stable, which shows the trend going down and no increase in vibration; the second trend is for when the machine is unstable, which shows the trend going up meaning the vibration is increasing and will continue to increase based on the current trend for the following second which can be used to adjust or stop the machine to reduce the chance of scraping the part.

7- PT_reconfiguration_service

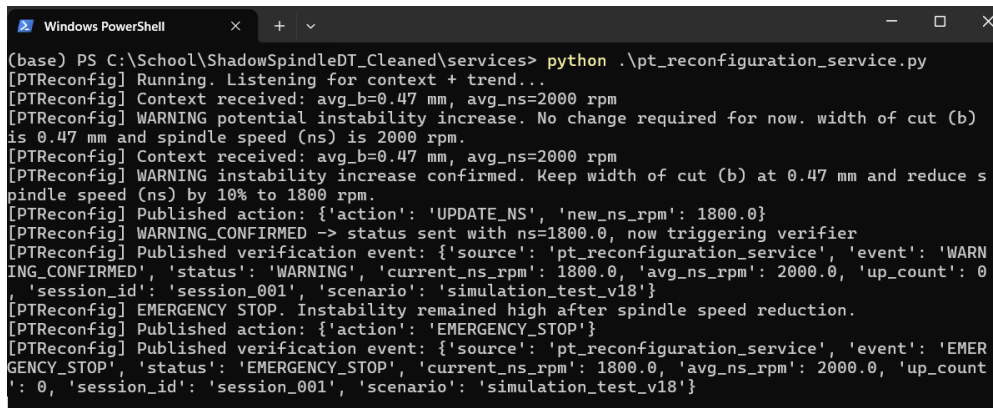
It is the main control-oriented service in architecture. It receives process-context information and stability-trend outputs, then decides how the system should respond. Depending on the detected condition, it can keep the process in a stable state, issue a warning, reduce spindle speed, or trigger an emergency stop. It also publishes status messages for the dashboard and visualization layers and emits verification events for downstream checking. This service is especially important because it serves as the primary feedback mechanism for the digital shadow. While much of the architecture is focused on observing and analyzing the process, the PT

reconfiguration service introduces action and intervention logic. Its value lies in translating analytical results into operational decisions, thereby connecting monitoring to response.



```
(base) PS C:\School\ShadowSpindleDT_Cleaned\services> python .\pt_reconfiguration_service.py
[PTReconfig] Running. Listening for context + trend...
[PTReconfig] Context received: avg_b=0.3 mm, avg_ns=2000 rpm
[PTReconfig] DOWN: No changes required. width of cut (b) is 0.3 mm and spindle speed (ns) is 2000 rpm.
[PTReconfig] Context received: avg_b=0.3 mm, avg_ns=2000 rpm
[PTReconfig] DOWN: No changes required. width of cut (b) is 0.3 mm and spindle speed (ns) is 2000 rpm.
[PTReconfig] Context received: avg_b=0.3 mm, avg_ns=2000 rpm
[PTReconfig] DOWN: No changes required. width of cut (b) is 0.3 mm and spindle speed (ns) is 2000 rpm.
[PTReconfig] Context received: avg_b=0.3 mm, avg_ns=2000 rpm
[PTReconfig] DOWN: No changes required. width of cut (b) is 0.3 mm and spindle speed (ns) is 2000 rpm.
```

Figure 9 PT_reconfiguration Microservice - Stable Pass

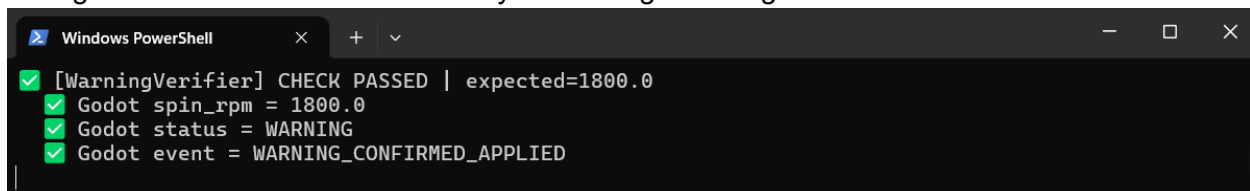


```
(base) PS C:\School\ShadowSpindleDT_Cleaned\services> python .\pt_reconfiguration_service.py
[PTReconfig] Running. Listening for context + trend...
[PTReconfig] Context received: avg_b=0.47 mm, avg_ns=2000 rpm
[PTReconfig] WARNING potential instability increase. No change required for now. width of cut (b)
is 0.47 mm and spindle speed (ns) is 2000 rpm.
[PTReconfig] Context received: avg_b=0.47 mm, avg_ns=2000 rpm
[PTReconfig] WARNING instability increase confirmed. Keep width of cut (b) at 0.47 mm and reduce s
pindle speed (ns) by 10% to 1800 rpm.
[PTReconfig] Published action: {'action': 'UPDATE_NS', 'new_ns_rpm': 1800.0}
[PTReconfig] WARNING_CONFIRMED -> status sent with ns=1800.0, now triggering verifier
[PTReconfig] Published verification event: {'source': 'pt_reconfiguration_service', 'event': 'WARN
ING_CONFIRMED', 'status': 'WARNING', 'current_ns_rpm': 1800.0, 'avg_ns_rpm': 2000.0, 'up_count': 0
, 'session_id': 'session_001', 'scenario': 'simulation_test_v18'}
[PTReconfig] EMERGENCY STOP. Instability remained high after spindle speed reduction.
[PTReconfig] Published action: {'action': 'EMERGENCY_STOP'}
[PTReconfig] Published verification event: {'source': 'pt_reconfiguration_service', 'event': 'EMER
GENCY_STOP', 'status': 'EMERGENCY_STOP', 'current_ns_rpm': 1800.0, 'avg_ns_rpm': 2000.0, 'up_count
': 0, 'session_id': 'session_001', 'scenario': 'simulation_test_v18'}
```

Figure 10 PT_reconfiguration Microservice - Unstable Pass

8- Warning_confirmed_state_formal_verification (Formal Verification 1)

It checks whether warning-confirmed actions are correctly reflected in the downstream system. It verifies that when the PT reconfiguration service confirms a warning and commands a spindle-speed adjustment, the expected updated state is observed in the visualization-related feedback path. It then publishes a verification result indicating whether the expected response was correctly applied. The value of this service is that it increases trust in the internal consistency of the digital shadow. It is not enough for a service to issue a command; the system must also verify that the intended effect is propagated through the pipeline. This verification layer strengthens confidence in the reliability of warning-handling behavior.



```
Windows PowerShell
[WarningVerifier] CHECK PASSED | expected=1800.0
[Godot] spin_rpm = 1800.0
[Godot] status = WARNING
[Godot] event = WARNING_CONFIRMED_APPLIED
```

Figure 11 Warning Microservice - Unstable Pass

9- Emergency_state_formal_verification (Formal Verification 2)

It performs a similar role for emergency-stop events. When an emergency stop is triggered, this service checks whether the downstream system, particularly the visualization state, reflects the expected emergency condition. If the expected state is not observed, it can reissue

the emergency-stop action and publish a failed verification result. This service is valuable because emergency-stop logic is one of the most safely critical parts of the project. The digital shadow must not only detect severe instability but also ensure that the emergency state is consistently applied across relevant system components. This verification service, therefore, supports robustness and trustworthiness in the most critical response scenario.

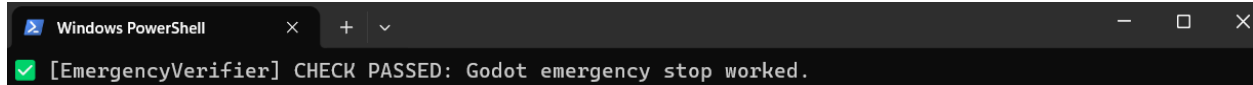


Figure 12 Emergency Microservice - Unstable Pass

10- Process_stop_formal_verification (Formal Verification 3)

This service is intended to verify that the physical twin has stopped when the process is completed. This happens when the process remains stable during its entirety. In practice, this verification concerns the consistency between control commands, producer behavior, and downstream system status after a stop event has been triggered. Its role is to confirm that the process no longer continues to publish or behave as if machining were still active once a stop decision has been made. The value of this verification function lies in increasing confidence that process-stop actions are not only generated at the decision layer but are also effectively enforced across the pipeline. This is especially important because stopping behavior is closely tied to safe operation and to the reliability of emergency and control-related responses in the digital shadow.

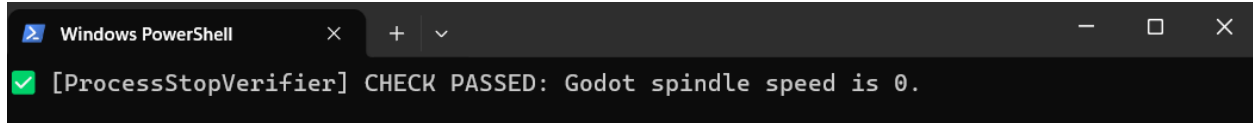


Figure 13 Process Verification Microservice

11- Rabbit_to_godot_bridge (Visualization Service)

It connects the RabbitMQ-based backend pipeline to the Godot visualization environment. It consumes movement, status, and context messages from RabbitMQ and forwards them through a WebSocket connection to Godot. It also receives state updates back from Godot and republishes them into RabbitMQ so that verification services can confirm that visual actions were applied correctly. This bridging role is important because the visualization environment does not directly consume RabbitMQ messages in the same way as the other backend services. The bridge, therefore, serves as a translation and communication layer between the analytics pipeline and the 3D simulation. Its value lies in enabling real-time visual representation of the machining process and its changing operating state.

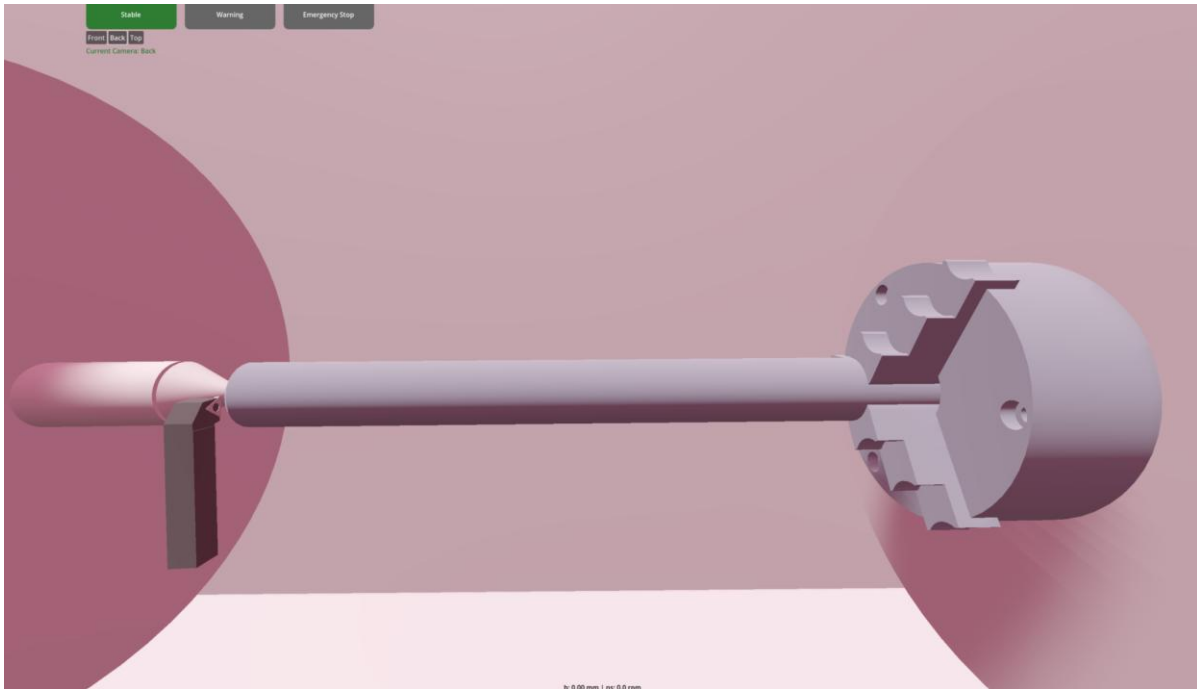


Figure 14 Visualization Service in Godot

12- Dashboard_consumer (Dashboard Service)

It provides the main user-facing monitoring interface for the project. It consumes live stream data, system status messages, verification results, and related analytical outputs, then presents them through an interactive dashboard built with Dash and Plotly. The dashboard displays live signals, FFT information, alarm-related indicators, status changes, and prediction-related views. This service is valuable because it transforms backend computations into interpretable insight for the user. The rest of the digital shadow may generate measurements, features, and decisions, but the dashboard is what makes those outputs accessible in practice. It plays a critical role in user interaction by allowing an operator or evaluator to observe the process state, understand emerging instability, and view the system's responses in real time. Together, these microservices form a coherent pipeline in which raw simulated process data is transformed into analytics, predictions, visualizations, verification results, and control-oriented decisions. This structure is one of the main strengths of the project, because it demonstrates how a digital shadow can be engineered as a modular and extensible software system rather than as a single tightly coupled application.

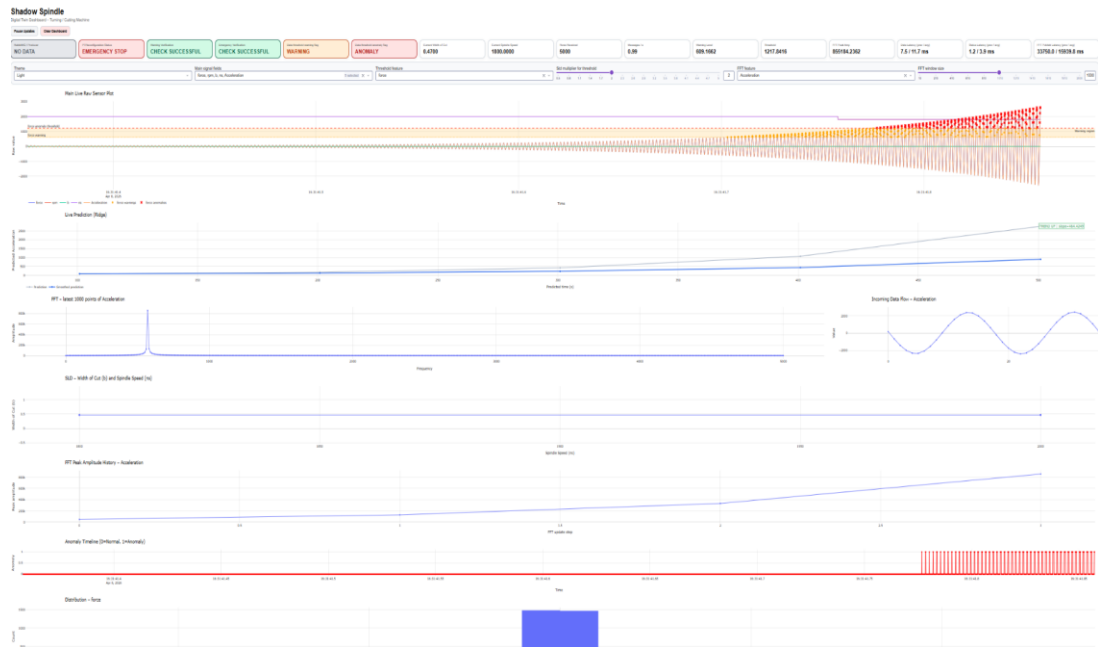


Figure 15 Dashboard Service in Python, Dash

This image shows the dashboard interface as the data is coming in which provides all the needed information to monitor the machine as it's working, it shows raw data plot, prediction of FFT peak amplitude, FFT graph, anomaly graph (threshold based).

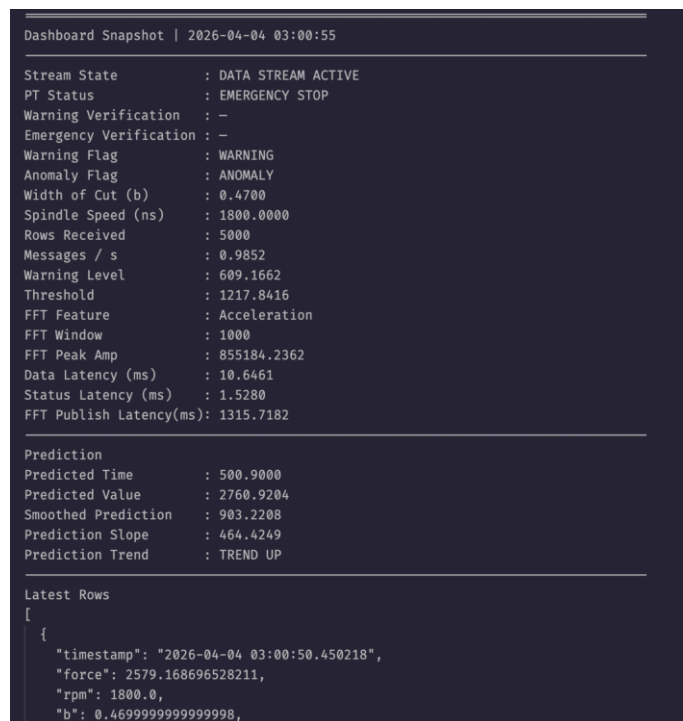


Figure 16 Snapshot of Dashboard Logs

This image shows the output of the dashboard to be used as a secondary close to the digital shadow to twin loop to support decision to be taken about the machine in case of failure or warning, this output is a log of all the information visible on the dashboard and the calculations done in real time logged as a snippet to be stored in the database for future reference and possible future prediction models built based on that information.

Standards

To relate the project to an established digital twin standard, the architecture was mapped onto the **ISO 23247** digital twin framework for manufacturing. This standard was chosen because it provides a structured reference architecture for manufacturing-oriented digital twins and helps organize the system into entities, sub-entities, and cross-system functions. The figure in this section adapts that framework to the Orthogonal Turning Digital Shadow. At the foundation of the mapping are the **Observable Manufacturing Elements (OMEs)**, which correspond to the physical process being represented: the orthogonal turning operation, including the lathe and spindle, workpiece, cutting tool, sensing context, and simulated process behavior. Above this is the **Data Collection and Device Control Entity**, which in this project includes signal acquisition and preparation, FFT-related preprocessing, process variables such as acceleration and spindle speed, and control-related actions such as spindle-speed adjustment and emergency stop.

The **Digital Twin Entity** contains the main digital services of the project. In the adapted mapping, this includes operation and management functions, analytical and decision-support services, and resource-access concerns. In practice, this layer covers the project's core software functionality, including simulation handling, stability analysis, prediction, PT reconfiguration, verification, and visualization support. Above it is the **User Entity**, represented mainly by the dashboard and Godot visualization, through which users observe system state and interpret machining behavior.

The figure also includes a **Cross-System Entity** for concerns that span the architecture, such as data translation, data assurance, and security. In the current project, RabbitMQ message exchange, session identifiers, timestamps, and consistency checks contribute to these functions, while security is represented conceptually but remains only partially developed. This mapping helps position the project within a recognized manufacturing digital twin framework while also highlighting what remains to be strengthened, especially real physical integration and broader cross-system concerns.

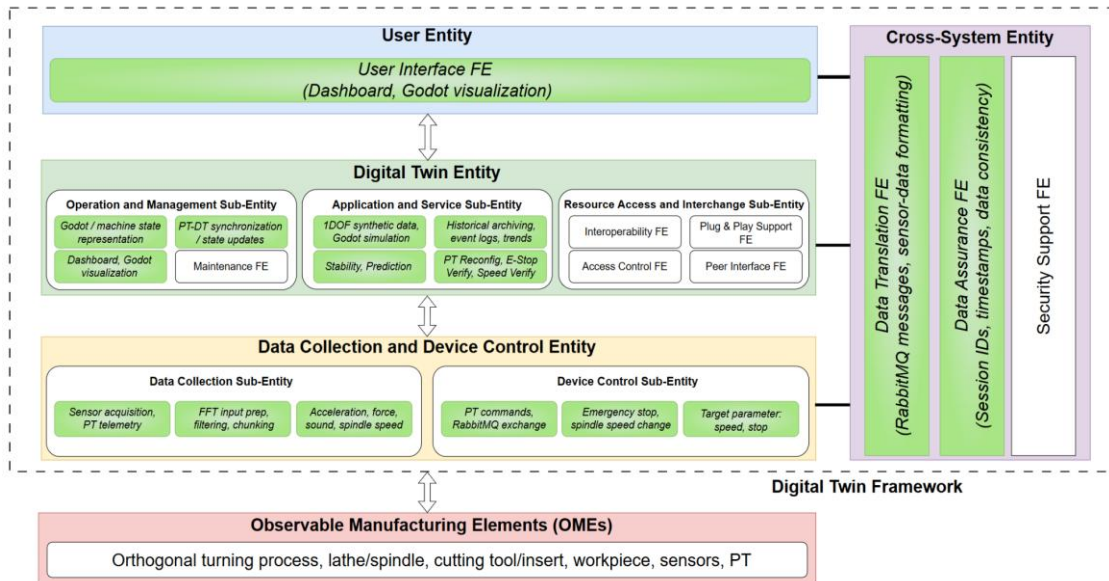


Figure 17 Digital Shadow Mapped to ISO 23247 Standards

Description Framework

To describe the project in a structured and standardized way, the Orthogonal Turning Digital Shadow was documented using the **DTInsight reporting** approach and the associated **Digital Twin Description Framework (DTDF)**. This framework was useful because it allowed the project to be represented not only as a software system, but also as a digital twin engineering artifact characterized through a common set of dimensions. The complete DTDF 21-characteristic table is provided in **Appendix A**, while the DTInsight reporting page generated during the project is available online.

The DTDF characteristics presented in **Appendix A** show that the project addresses the main technical foundations of a manufacturing-oriented digital shadow. The system under study is clearly defined as an orthogonal turning process involving a lathe, workpiece, cutting tool, and sensing context. The project includes physical sensing concepts, physical-to-virtual interaction, virtual services, near-real-time operation, and a one-to-one relationship between the represented process and its digital counterpart. It also combines physics-based and data-driven elements, including synthetic simulation data, FFT-based signal analysis, stability assessment, prediction, and reconfiguration logic. Tooling and enablers such as RabbitMQ, Python, Godot, InfluxDB, and Dash support the implementation. At the same time, the framework highlights current limitations: the technical connection to the physical process remains incomplete because the implementation still relies mainly on simulated data, while standardization and security are only addressed at an initial level.

The constellation diagram complements the characteristic table by showing how the physical process, digital services, models, data, and enabling technologies are related. This makes it useful for presenting the project as a structured ecosystem rather than only as a sequence of software services. The completed DTInsight reporting page is available at:

<https://haitham-alsatai.github.io/DTDF-Reporting/>.

Constellation Diagram

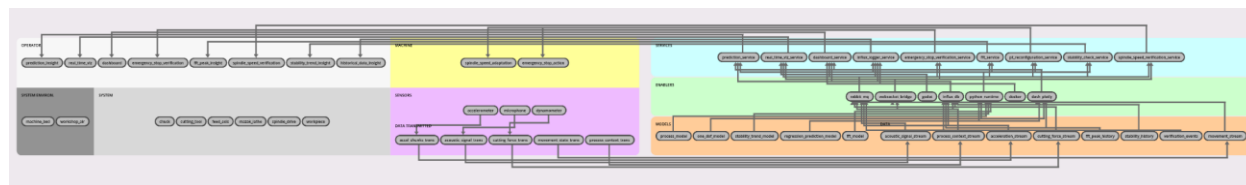


Figure 18 Constellation Diagram

Conclusion

This report presented the design and implementation of an Orthogonal Turning Digital Shadow for monitoring machining stability during cutting operations. Using synthetic data generated from a 1DOF model, the project integrated microservices for data production, FFT-based signal analysis, stability assessment, prediction, reconfiguration, logging, dashboard reporting, formal verification, and 3D visualization. The resulting system demonstrates that a modular digital shadow can provide near-real-time insight into process behavior and support corrective reasoning in a structured and scalable way. According to Kritzinger's framework, the current system is best classified as a digital shadow because it includes automatic process-to-virtual data flow, but it does not yet achieve fully validated bidirectional integration with the real physical machine. The following sections discuss the main challenges encountered, the lessons learned, the current limitations of the implementation, and the future work required to move the project closer to a complete digital twin.

Challenges

Throughout the duration of this project many challenges were incurred. In this section, a total of three of the main challenges experienced will be explained in greater detail. This challenge was regarding obtaining synthetic data. Since this project is not connected to a real physical twin, the team had to generate synthetic data that was representative of the system. To mitigate this challenge, the team met with a domain expert and a one degree of freedom model was created. The second main challenge experienced during the project was determining the correct architecture for this digital shadow solution and how to connect everything together using communicators such as RabbitMQ. Microservices were being constructed individually but the team took a lot of time before connecting everything together, adding consumer and producer components to all the microservices and that caused a lot of reworks for the team. The final challenge lies in the inherently multidisciplinary nature of the digital twin field, which demands collaboration across diverse areas of expertise. It was extremely important to maintain an agile methodology and present our progress to domain experts to make sure the evolution of the solution was aligned with their needs.

Lessons Learned

To continue the multidisciplinary aspect of the digital twin field, this was one of the most important lessons learned during the past four months. It happened during the project duration that we demonstrated progress and the feedback received was crucial and led to future changes such as modifying the spindle speed instead of the width of the cut or adding cameras in our visualization service. Another lesson learned would have been to incorporate the ISO 23247 standards earlier in the design phase. This would help guide the architecture and brainstorming sessions. Finally, the last lesson learned is that most of the team members had no prior experience in tools such as RabbitMQ, Godot, influx DB and more, and the team had to do research and learn how to use all the tools available to us to develop a solution.

Limitations

In terms of limitation, this project currently only functions for a specific case of orthogonal turning. It is also simplified, assuming that the model dynamics are fixed and do not evolve through time. Furthermore, the current digital shadow does not consider resonance, which is a frequency that generates higher vibrations, but still is a stable vibration and so future work will include incorporating a resonance detector and won't change the state of the system. On another note, the current digital shadow solution has a prediction service that acts like a monitoring service one second ahead of the current time. The team prioritized having a prediction service that worked and could be used to monitor the system one second in advance.

Future Work

Finally, future work opportunities have been identified for the coming months. For instance, in relation to the prediction service, one future work identified is improving the prediction service and allowing the user to select between the real-time stability service trend identification or the prediction service to communicate with the PT_reconfiguration service. Another future work planned for the upcoming months is to connect the architecture with the physical twin and replace the synthetic data with real machine data. Finally, the last future work opportunity lies in generalizing the architecture with the hope of being able to extend this architecture with other processes than orthogonal cutting.

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- [3] "What is chatter in machining and how to reduce it," ScienceInsights, <https://scienceinsights.org/what-is-chatter-in-machining-and-how-to-reduce-it/> (accessed Apr. 13, 2026).
- [4] GeeksforGeeks, "What are the advantages and disadvantages of microservices architecture?," GeeksforGeeks, <https://www.geeksforgeeks.org/system-design/what-are-the-advantages-and-disadvantages-of-microservices-architecture/> (accessed Apr. 13, 2026).

Appendix A: 21 Characteristics

<u>Characteristics</u>	<u>Does our DT have it?</u>	<u>Description</u>
C1: System-Under-Study	Yes	- Orthogonal turning process. - Lathe (chuck), pipe (AL 7075), insert (cutting tool) - Machine: Mazak QT-Nexus-200
C2: Physical Acting Components	Yes	The DT will be able to control the spindle speed, and if the system enters an unstable state, it will be able to stop the lateral movement of the tool.
C3: Physical Sensing Components	Yes	Accelerometer, dynamometer, microphone.
C4: Physical-to-Virtual Interaction	Yes	Displacement, force, acceleration, sound
C5: Virtual-to-Physical Interaction	Yes	Messages regarding what to do next with the spindle speed (keep same of change). If the system enters an unstable state, then the DT will trigger the emergency stop state and stop everything.
C6: DT Services	Yes	Services - real-time visualization - simulation service - stability service - PT Reconfiguration service - Prediction service
C7: Twinning Timescale	Yes	This DT operates near real-time taking a window of multiple points and the entire cycle runs in X amount of time.
C8: Multiplicities	1 to 1	1 DT for 1 PT
C9: Life-Cycle Stages	Yes	Purpose of DT: operate the PT
C10: DT Models in Data	Yes	We are using physics-based models and data-driven models in this DT. - physics-based model: FFT calculation, 1DOF to represent the system. - data-driven model: stability trends
C11: Tooling and Enablers	Yes	InfluxDB, RabbitMQ, Godo, pythont enable us to visually communicate data through the PT and DT.
C12: DT Constellation	Yes	Refer to the constellation diagram later in this report.
C13: Twinning Process and DT Evolution	Yes	Engineering steps: - Meet with domain experts and understand their needs and challenges. - Create the 1DOF model and obtain synthetic data. - Determined the architecture of the

		microservices and the contents of each microservices • Create microservice architecture • Obtain feedback from SME (subject matter expert) and begin the validation loop as an interactive process.
C14: Fidelity and Validity Consideration	Yes	The current DT is connected to synthetic data of a 1 degree of freedom system (1DOF) which has no noise and basically is perfect (perfectly stable or perfectly unstable). Eventually this DT will be connected to real data, coming from a lathe performing an orthogonal cut. When we are there, we will be able to ensure the fidelity and validity of DT. For now, as a form of validity we are meeting with domain experts and presenting our work to get feedback and comments on how to improve the DT architecture and services to ensure the domain experts will want to use it later.
C15: DT Technical Connection	No	DT is currently not connected with a PT. Connecting it is the next step. We have the machine; we ordered the parts and once received we can begin performing experimental runs.
C16: DT Hosting and Deployment	Yes	This project has a GitHub repository where the digital twin exists, we then run everything locally.
C17: Insights and Decision Making	Yes	This DT will control cutting parameters, mainly spindle speed, to ensure the system remains within stability bounds. This DT also has a dashboard that can be used to gain insight into the current and past (and future with prediction service) system.
C18: Horizontal Integration	N/A	Not applicable because the lathe is not connected to any other machines.
C19: Data Ownership and Privacy	N/A	Not considered in this study.
C20: Standardizations	No	Not currently. Short term work: map this DT to ISO standards explained in Ishvan's lecture. Future work: This DT will be transformed into a scalable and generalized way that will allow standardized methods to be developed and used by other domain experts.
C21: Security and safety considerations	Yes	This DS eventually DT will make the environment safer by carefully monitoring the system and making sure it stays within stability bounds. Short-term work: add spindle speed ranges based on the fact we know we are working with AL 7050.

		Future work: add tables presenting what the DT can control and what the DT can control but requires human input.
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